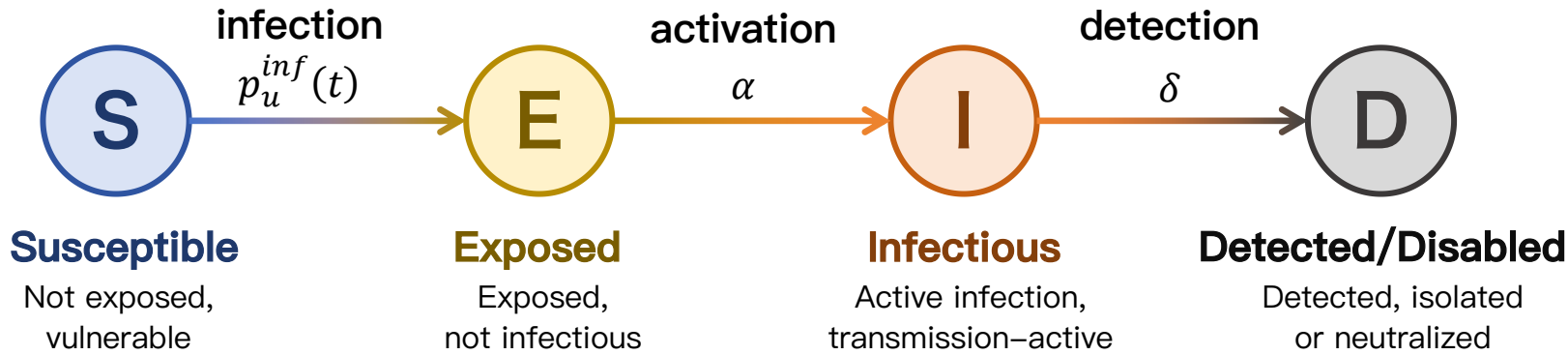
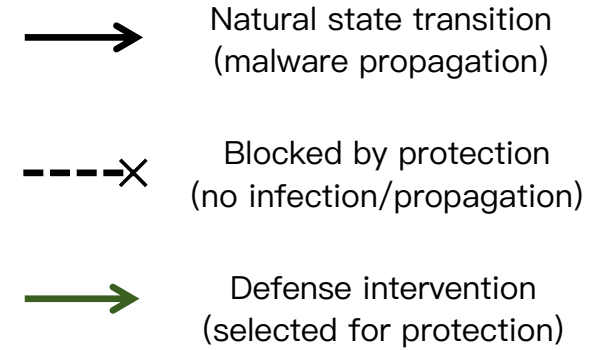


Natural malware propagation (SEID chain)



Legend



Defense intervention at time t

Defender selects a protected node set $\mathcal{U}(t) \subseteq \mathcal{V}^c$
(budget constraint: $|\mathcal{U}(t)| \leq B_d$)

Eligible current states

The selected node can be in any of the following states at time t



selected
($u \in \mathcal{U}(t)$)

Protected

Protected by defense;
excluded from infection
and propagation within current
defense horizon

Effect of protected state P

- **Cannot be infected**
($\Pr[x_u(t+1) = E | x_u(t) = P] = 0$)
- **Cannot transmit malware to neighbors**
- **Excluded from infection and propagation within current defense horizon**

Notation: $x_u(t)$: state of node u at time t $\mathcal{U}(t)$: protected node set at time t $\mathcal{V}^c$: set of communication nodes B_d : defense budget